Jacobian Descent For Multi-Objective Optimization

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Joint work with Pierre Quinton

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GitHub
github.com/TorchJD/torchjd

TorchJD
torchjd.org

arXiv
arxiv.org/pdf/2406.16232

Table of contents

Theory

Multi-objective optimization

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, as

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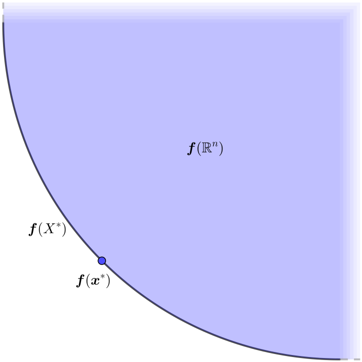

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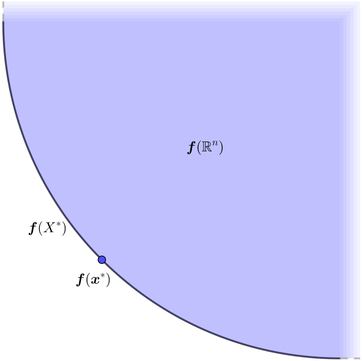

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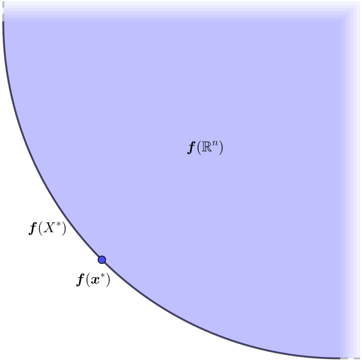
Pareto Front: $\boldsymbol{f}(X^*)$



First-order Taylor approximation

For any $\boldsymbol{x} \in \mathbb{R}^n$, the Jacobian of $\boldsymbol{f}$ at $\boldsymbol{x}$, is

$$\mathcal{J}\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \nabla f_1(\boldsymbol{x})^\top \\ \nabla f_2(\boldsymbol{x})^\top \\ \vdots \\ \nabla f_m(\boldsymbol{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\boldsymbol{x}) \\ \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_2(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\boldsymbol{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_m(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\boldsymbol{x}) \end{bmatrix}$$

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For a small enough update $\boldsymbol{y} \in \mathbb{R}^n$, Taylor's theorem states

$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

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We select $\boldsymbol{y} = -\eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x})) \in \mathbb{R}^n$.

Jacobian descent

The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

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Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

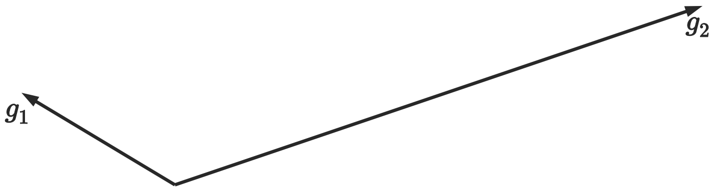
Input: $\boldsymbol{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$

for $t \leftarrow 1$ **to** T **do**

  $\boldsymbol{x} \leftarrow \boldsymbol{x} - \eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$

Output: $\boldsymbol{x}$

Unconflicting projection of gradients (UPGrad)



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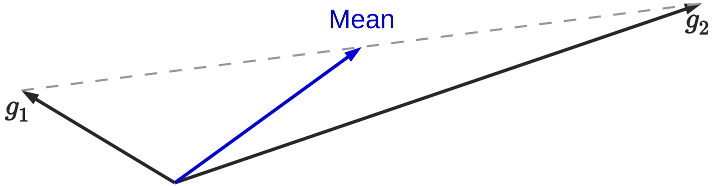

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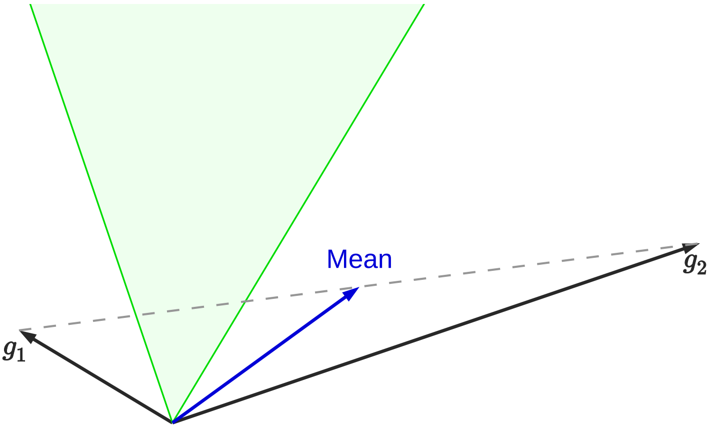

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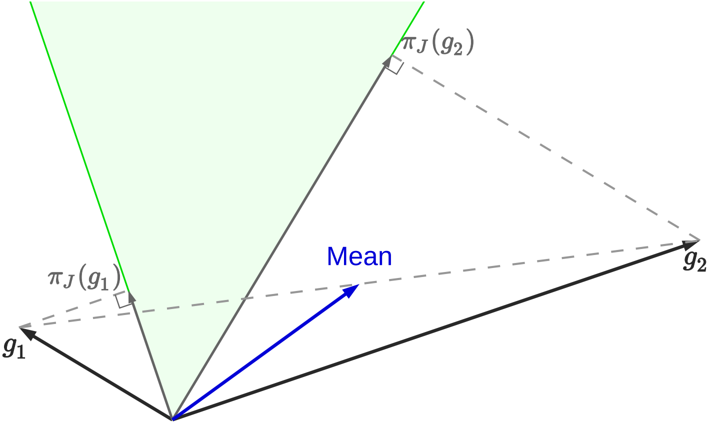

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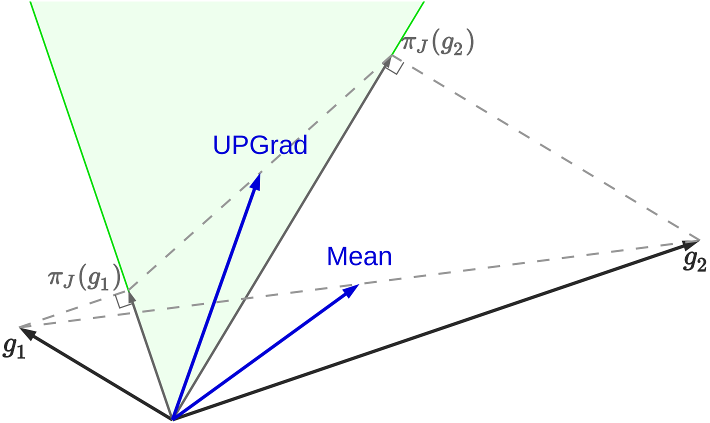


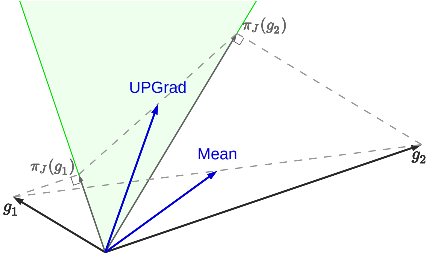
Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let C_J be the cone $\{\boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{0} \preceq J\boldsymbol{y}\}$. $\mathcal{A}$ is non-conflicting if for all $J \in \mathbb{R}^{m \times n}$, $\mathcal{A}(J) \in C_J$.

For any $\boldsymbol{x} \in \mathbb{R}^n$, the projection of $\boldsymbol{x}$ onto C_J is

$$\pi_J(\boldsymbol{x}) = \underset{\boldsymbol{y} \in C_J}{\arg\min} \|\boldsymbol{x} - \boldsymbol{y}\|^2$$

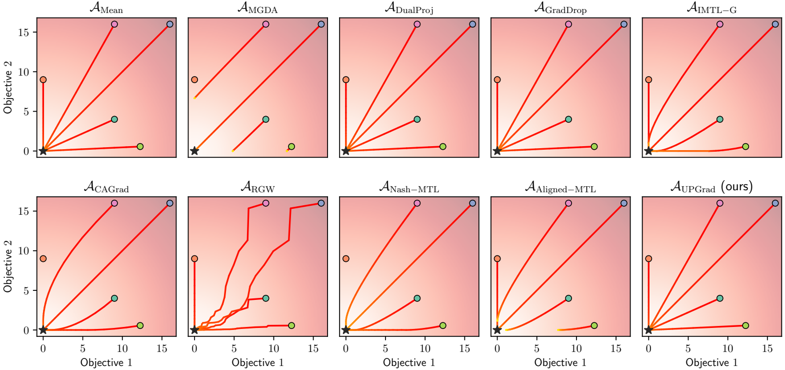
$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} \pi_J\left(J^\top \boldsymbol{e}_i\right)$$



Applications and experimentation

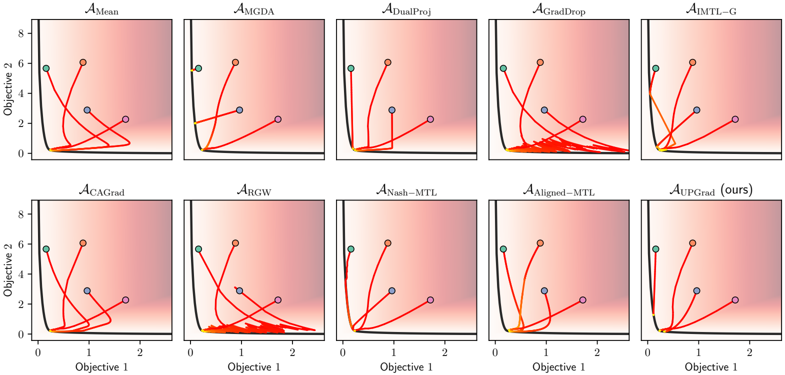
Synthetic example: element-wise quadratic function

Minimize $\boldsymbol{f} : \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$. It's Jacobian at $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$.

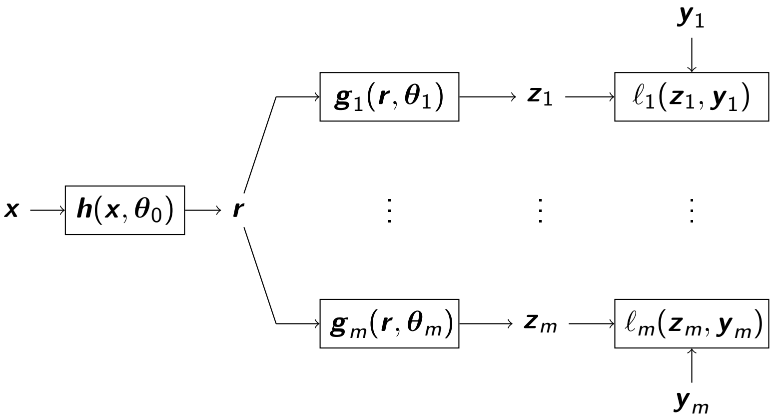


Synthetic example: convex quadratic form

Some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with highly conflicting gradients.



Multi-task learning



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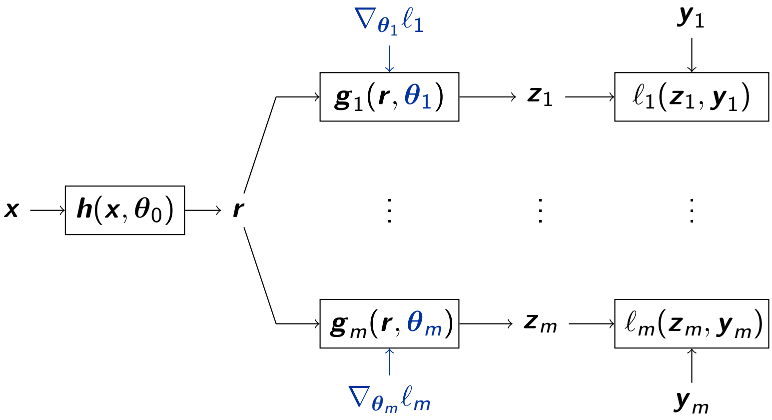

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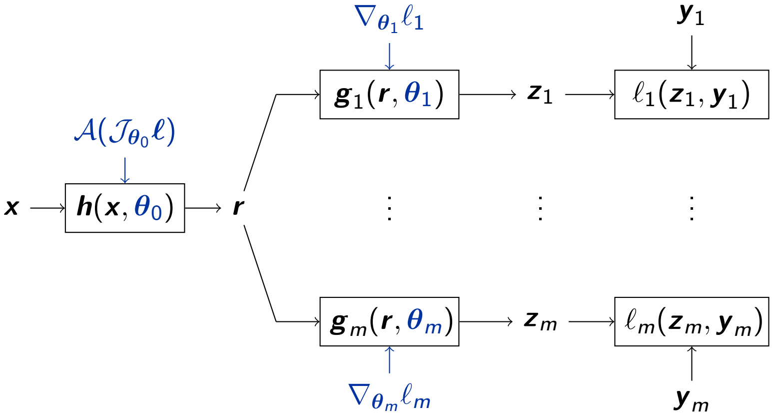

Experimental results: per-sample JD

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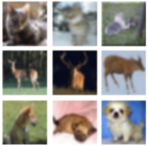
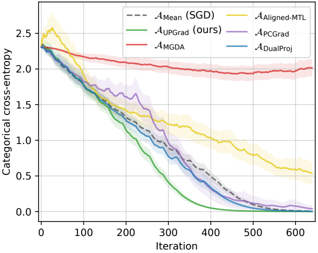
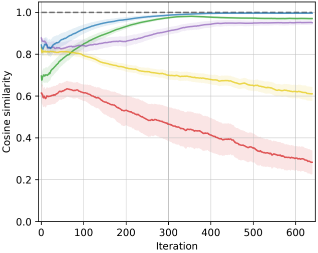
CIFAR-10



Training Loss



Update Similarity



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- Find new applications.

My research interests

Finding new ways to train neural networks.

Finding new architectures (that are hard to train with SGD).

Making algorithms faster (e.g. new custom tensors, etc.).

Contributing to the open-source community.

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Applicable to most other aggregators from the literature.